# Porphyria Cutanea Tarda (PCT)

Porphyria Cutanea Tarda (PCT) is the most common of the Porphyrias and results from a deficiency of the enzyme uroporphyrinogen decarboxylase (UROD). Porphyria Cutanea Tarda (PCT) is a rare disorder characterized by painful, blistering skin lesions that develop on sun-exposed skin (photosensitivity). Affected skin is fragile and may peel or blister after minor trauma. Liver abnormalities may also occur.

PCT is essentially an acquired disease, but some individuals have a genetic (autosomal dominant) deficiency of UROD that contributes to development of PCT. These individuals are referred to as having "familial PCT". Most individuals with the inherited enzyme deficiency remain latent and never have symptoms.

Large amounts of porphyrins build up in the liver when PCT is becoming active. The disease becomes active when acquired factors, such as iron, alcohol, Hepatitis C Virus (HCV), HIV, estrogens (used, for example, in oral contraceptives and prostate cancer treatment) and possibly smoking, combine to cause a deficiency of UROD in the liver. Hemochromatosis, an iron overload disorder, also can predispose individuals to PCT.

In patients with acquired PCT, males and females are affected equally, but in patients with familial, genetic PCT, females are affected more than males. The disorder usually develops after the age of 30 and its onset in childhood is rare. PCT is found worldwide and in individuals of all races. PCT is a rare disorder; the prevalence is estimated to be approximately 1 case in every 10,000 to 25,000 individuals in the general population.

### Synonyms of Porphyria Cutanea Tarda

- UROD deficiency
- uroporphyrinogen decarboxylase deficiency

### Subdivisions of Porphyria Cutanea Tarda

- sporadic or acquired Porphyria Cutanea Tarda (PCT type 1)
- familial Porphyria Cutanea Tarda or f-PCT (PCT type 2)

## Cause

Porphyria Cutanea Tarda is caused by deficient levels of an enzyme known as uroporphyrinogen decarboxylase (UROD). In approximately 75% to 80% of cases this deficiency is acquired (PCT type 1 or sporadic PCT); in the remaining cases, individuals have a genetic predisposition to developing the disorder, specifically a mutation in the UROD gene (PCT type 2 or familial PCT). Most individuals with this genetic mutation do not develop symptomatic PCT; the mutation is a predisposing factor and additional factors, known as triggers, are required for the development of the disorder in these individuals. These factors are called susceptibility factors and are required for the development of both sporadic and familial PCT. Generally, PCT develops in mid to late adulthood. In extremely rare cases, individuals have mutations in both UROD genes. This autosomal recessive form of familial PCT is known as Hepatoerythropoietic Porphyria (HEP). HEP occurs in childhood and is usually more severe than PCT types 1 or 2. NORD has a separate report on HEP.

PCT is a multifactorial disorder, which means that several different factors such as genetic and environmental factors occurring in combination are necessary for the development of the disorder. These factors are not necessarily the same for each individual. These factors contribute either directly or indirectly to decreased levels or ineffectiveness of an enzyme known as uroporphyrinogen decarboxylase (UROD) within the liver. When UROD levels in the liver decrease to approximately 20% of normal levels, the symptoms of PCT may develop.

The UROD enzyme is essential for breaking down (metabolizing) certain chemicals in the body known as porphyrins. Low levels of functional UROD result in the abnormal accumulation of specific porphyrins in body, especially within the blood, liver and skin. The symptoms of PCT occur because of this abnormal accumulation of porphyrins and related chemicals. For example, when porphyrins accumulate in the skin, they absorb sunlight and enter an excited state (photoactivation). This abnormal activation results in the characteristic damage to the skin found in individuals with PCT. The liver removes porphyrins from the blood plasma and secretes it into the bile. When porphyrins accumulate in the liver, they can cause toxic damage to the liver.

The exact, underlying mechanisms that cause PCT are complex and varied. It is determined that iron accumulation within the liver plays a central role in the development of the disorder in most individuals. Recently, researchers have discovered that a substance called uroporphomethene, which is an oxidized form of a specific porphyrin known as uroporphyrinogen, is an inhibitor that reduces the activity of the UROD enzyme in the liver. The oxidation of uroporphyrinogen into uroporphomethene has been shown to be iron dependent, emphasizing the importance of elevated iron levels in the development of PCT.

The relationship between iron levels and PCT has long been established and PCT is classified as an iron-dependent disease. Clinical symptoms often correlate with abnormally elevated levels of iron in the liver (iron overloading). Iron overloading in the liver may only be mild or moderate. The exact relationship between iron accumulation and PCT is not fully understood, as there is no specific level of iron in the liver that correlates to disease activity in PCT (e.g. some individuals with symptomatic PCT have normal iron levels).

There is an increased prevalence of mutations in the HFE gene in individuals with PCT. Mutations in the HFE gene can cause hemochromatosis, a disorder characterized by the accumulation of iron in the body, especially the liver. Hemochromatosis occurs when a person inherited two mutated HFE genes (one from each parent). Hemochromatosis is associated with low levels of hepcidin, a specialized protein that is the primary regulator of iron absorption in the body, including regulating the uptake of iron by the gastrointestinal tract and liver.

Additional risk factors that have been associated with PCT include alcohol, certain infections such as hepatitis C or HIV, and drugs such as estrogens. Some studies have indicated that smoking is a risk factor for PCT in susceptible individuals. Less often, certain chemical exposures (e.g. hexachlorobenzene), kidney dialysis, and lupus appear to be connected to the development of PCT. It is believed that these susceptibility factors reduce hepcidin in the body and consequently lead to iron accumulation in the liver. However, the exact relationship among most susceptibility factors with the development of symptoms in PCT is not fully understood. For example, alcohol clearly contributes to the development of the disorder in some cases, but PCT is not common in alcoholics. Most individuals with PCT have three or more susceptibility factors present.

The underlying cause of UROD deficiency in the acquired form of PCT is unknown. Affected individuals have approximately 50% residual UROD activity and do not develop symptoms unless additional factors are present. The most common factors associated with acquired PCT are hemochromatosis or chronic hepatitis C infection. In individuals with acquired PCT, UROD levels are only deficient in the liver.

In the familial form of PCT, individuals have a mutation in the UROD gene. This mutation is inherited as an autosomal dominant trait. Genetic diseases are determined by the combination of genes for a particular trait that are on the chromosomes received from the father and the mother. Dominant genetic disorders occur when only a single copy of an abnormal gene is necessary for the appearance of the disease. The abnormal gene can be inherited from either parent or can be the result of a new (de novo) mutation in the affected individual with no family history. The risk of passing the abnormal gene from affected parent to offspring is 50% for each pregnancy regardless of the sex of the resulting child.

The UROD gene creates (encodes) the UROD enzyme, which is the fifth enzyme in the heme synthesis pathway. A mutation in this gene leads to abnormally low levels of this enzyme in all tissues of the body (not just the liver). However, one mutation alone is insufficient to cause familial PCT as residual UROD enzyme

levels remain above 20% of normal. In fact, most individuals with a mutation in the UROD gene do not develop the disorder. Additional factors must be present for the disorder to develop.

## Signs and Symptoms

Skin abnormalities characterize this disorder. Affected individuals are particularly susceptible to damage of the skin from sunlight (photosensitivity). Extremely fragile skin that can peel or blister on minimal impact is common. Affected individuals may develop blistering skin lesions on areas of the skin that are frequently exposed to the sun such as the hands and face. These lesions may crust over. Eventually, scarring may develop, and affected skin may darken (hyperpigmentation) or fade (hypopigmentation) in color. Abnormal, excessive hair growth (hypertrichosis), especially on the face may also occur. The hair may be very fine or coarse and can differ in color. In some patients, their hair may grow, thicken and darken. Small bumps with a distinct white head (milia) may also develop, especially on the backs of the hands.

Liver abnormalities may develop in some affected individuals including the accumulation of iron in the liver (hepatic siderosis), the accumulation of fat in the liver (steatosis), inflammation of certain parts of the liver (portal triaditis) and thickening and scarring around the portal vein (periportal fibrosis). Affected individuals may be at a greater risk than the general population of developing scarring of the liver (cirrhosis) or liver cancer known as hepatocellular carcinoma. Advanced liver disease is uncommon, except in older individuals with recurrent disease. In some cases, liver disease is due to an associated condition such as hepatitis C infection.

The symptoms of PCT can vary greatly from one individual to another.

## Diagnosis

The preferred screening test for PCT is a measurement of porphyrins in plasma. This can differentiate PCT from Variegate Porphyria. The patterns of porphyrins in urine (predominately uroporphyrin and 7-carboxylate porphyrin) and feces (predominately isocoproporphyrin) help to confirm the diagnosis. The presence of an inherited deficiency of UROD can be demonstrated by measuring the enzyme in red blood cells and is present in about 20% of patients with PCT.

A diagnosis of PCT is based upon identification of characteristic symptoms, a detailed patient history, a thorough clinical evaluation, and a variety of specialized tests.

## Treatment

PCT is the most treatable form of Porphyria and treatment appears equally effective for both the sporadic and familial forms. The standard treatment of individuals with PCT is regularly scheduled phlebotomies to reduce iron and porphyrin levels in the liver. This is the preferred treatment of affected individuals at many Porphyria centers regardless of whether there is confirmed iron overload. A phlebotomy is a simple and safe procedure that involves removing blood via a vein (bloodletting). Since much of the iron in the body is present in red blood cells, regular phlebotomies can reduce excess iron levels in the body. Regularly scheduled phlebotomies usually result in complete remission in most individuals. A phlebotomy schedule is recommended to achieve a target ferritin level of less than 20 nanograms per milliliter (<20 ng/mL). Ferritin is an iron compound that is used an indicator of the body's iron stores. Most patients require between five and eight phlebotomies to achieve remission.

In some cases, affected individuals may be treated with low doses of chloroquine and hydroxychloroquine, which can also reduce iron levels in the liver. These drugs are often used to treat malaria (antimalarials). This therapy is usually reserved for individuals for whom phlebotomies are not an option (e.g. contraindicated) such as in individuals with anemia; if there is the non-availability of venous access, or because of patient choice. The dosage of these drugs is especially important; dosages approaching those commonly used to treat individuals with other conditions can cause significant adverse effects in individuals with PCT including

elevating porphyrin levels and worsening photosensitivity. The recommended dosages are 100 mg twice a week for hydroxychloroquine or 125 mg twice a week for chloroquine. Such a low dose schedule is equally effective as phlebotomy and easier to take with less treatment cost involved. The mechanism of action of these drugs in individuals with PCT is not fully understood, but it is speculated that these drugs bind with porphyrins inside the lysosomes of liver cells, to be eventually excreted in the urine.

Hydroxychloroquine and chloroquine are contraindicated in pregnant women or women who are lactating. These drugs are also contraindicated for individuals with advanced liver disease, psoriasis, retinal disease, or glucose-6-phosphate dehydrogenase deficiency or who have recent or continued use of alcohol or drugs that are toxic to the liver (e.g. acetaminophen, isoniazid or valproic acid). Hydroxychloroquine and chloroquine can be associated with side effects including less serious ones (e.g. nausea, vomiting, headaches, etc.), but also more serious ones including seizures, muscle weakness or damage to the retinas of the eyes (retinopathy). Although retinopathy is unlikely with the low dose regimen used for PCT, an eye (ophthalmological) examination is recommended both before and after treatment. Signs of retinopathy can include blurred vision, light sensitivity or seeing halos around lights.

Iron chelators are drugs that bind to iron in the body allowing iron to be dissolved in water and excreted from the body through the kidneys. Iron chelators are less effective than phlebotomy or low dose hydroxychloroquine or chloroquine in treating individuals with PCT. However, these drugs may play a role in treating affected individuals in whom the use of the two front-line therapies is not possible, such as individuals with end stage renal disease who are on hemodialysis.

Affected individuals are advised to avoid environmental triggering factors of the disorder such as alcohol and smoking. The avoidance of sunlight may be necessary to protect the skin and can include the use of double layers of clothing, long sleeves, wide brimmed hats, gloves, and sunglasses. Pain killers (oral analgesics) can be used to treat painful skin lesion. Care should be taken to avoid infection of skin lesions. Antibiotics can be used to treat skin infections that do develop.

The treatment of PCT can achieve complete remission in affected individuals, but relapse is possible. The treatment of relapse is the same as the initial treatment.

## Investigational Therapies

Information on current clinical trials is posted on the Internet at www.clinicaltrials.gov. All studies receiving U.S. government funding, and some supported by private industry, are posted on this government web site.

## PCT and Hepatitis C Virus

Because PCT is frequently associated with Hepatitis C Virus (HCV) infection, it is worth noting the issues involved in treating a patient with both PCT and HCV infection.

Infection with HCV is much more common than PCT, and most people with HCV do not have PCT. However, at least in some locations, as many as 80 percent of individuals with PCT are infected with HCV. Therefore, HCV needs to be added to the list of factors that can activate PCT alongside alcohol, iron and estrogens. Other hepatitis viruses are seldom implicated in PCT, and it is not known how HCV activates PCT.

There are several different viruses that cause hepatitis. A blood test for HCV infection has not been available for very long. HCV is most readily transmitted from one person to another by blood products. Although most people who are infected with HCV have a history of exposure to blood or needles contaminated with blood, in some cases it is not known how the infection was acquired. HCV (unlike the Hepatitis B Virus and HIV) is seldom transmitted by sexual contact. It is also not readily transmitted by casual contact with other people. Therefore, people infected with HCV are not hazardous unless they somehow expose others to their blood.

It is recommended that patients with PCT be tested for HCV infection. This is done by a blood test that detects antibodies to the virus. If HCV infection is found, it may not change the treatment of PCT (by phlebotomy or low-dose chloroquine). Treatment for PCT is highly successful even in patients with HCV.

Therefore, it is reasonable to treat the PCT first and then look into treatment for HCV later.

There are reasons not to treat the HCV infection before treating the PCT. HCV treatment with alpha-interferon and ribavirin is available but is often not effective. Also, liver damage progresses slowly if at all in many people with HCV. However, once the PCT is in remission it is important to assess the amount of liver damage the virus has already caused and to have follow-up visits to a doctor to monitor the liver. In some cases, it may be important to treat HCV infection to try and prevent progressive liver damage.

**See Member Stories on Porphyria Cutanea Tarda (PCT)**

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